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Color Statistical Mapping by the U.S. Bureau of the Census

Morton A. Meyer, Frederick R. Broome, and Richard H. Schweitzer, Jr.

ABSTRACT. In recent years the U.S. Bureau of the Census has produced a number of thematic maps in color. These maps, appearing in various publication series, utilize predominantly the choropleth method. A variation in the use of this method is the two-variable "cross" map on which the distributions are mixed to represent category combinations. An important technological development that has enabled the production of color maps, specifically those in the Urban Atlases, has been the use of computer output on microfilm.

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Introduction

The dominant influence on mapmaking at the U.S. Bureau of the Census has been the operational needs of the organization's data collection activities. The majority of the Bureau's mapping activity is directed toward the development and updating of maps to be used in census field operations. Mapmaking for small-scale display of socio-economic data has generally been limited to the periods between major censuses. Even then, the first priority is the production of maps published in the volumes of tabular data. This paper reports on recent Bureau efforts to expand both the number and types of color thematic maps of census data.

The GE–50 and 70 Series

The first major activity in color statistical cartography in recent years has been the preparation of the GE–50 (Geography Series 50) maps.[1] The series, initiated in the early 1960's, is intended for either wall display or desk use. Each map is 106 by 76 cm. (42 by 30 in.). They are printed at a scale of 1:5,000,000 and depict distributions in one of three categories: (1) people and their economic characteristics; (2) socio-economic measures, such as percent of population in poverty; or (3) area boundaries of major interest, such as congressional districts. The series now includes over 60 maps based on various census data of which 14 show 1970 Decennial Census data (see appendix). The majority are choropleth maps. A much reduced version of a recent GE–50 series map, depicting Families Below the Low-Income Level in 1969, is shown in Fig. 1.

In the production of the GE–50 series, various color schemes have been used in an effort to develop effective arrangements of colors. Figure 2 shows some of the color schemes selected for use on published maps. The arrangement in Fig. 2–A is based on the use of the whole spectrum, with yellow representing the average value for the data while the colors for the higher values increase in blueness and the colors for the lower values increase in redness. Figure 2–B is a similar scheme but has more intermediate tones. The scheme in Fig. 2–C uses yellow for the lowest values, then shades of green, blue, and purple for the higher values. The colors in Fig. 2–D progress from pale yellow for the low data values, through oranges and pinks for middle values, to red for the highest values, while the scheme in Fig. 2–E progresses from pale yellow for low values through greens to dark blue for high values. Figure 2–F has two hues, reds representing decrease and blues representing increase. Throughout the map

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The American Cartographer, Vol. 2, No. 2, 1975, pp. 100–117

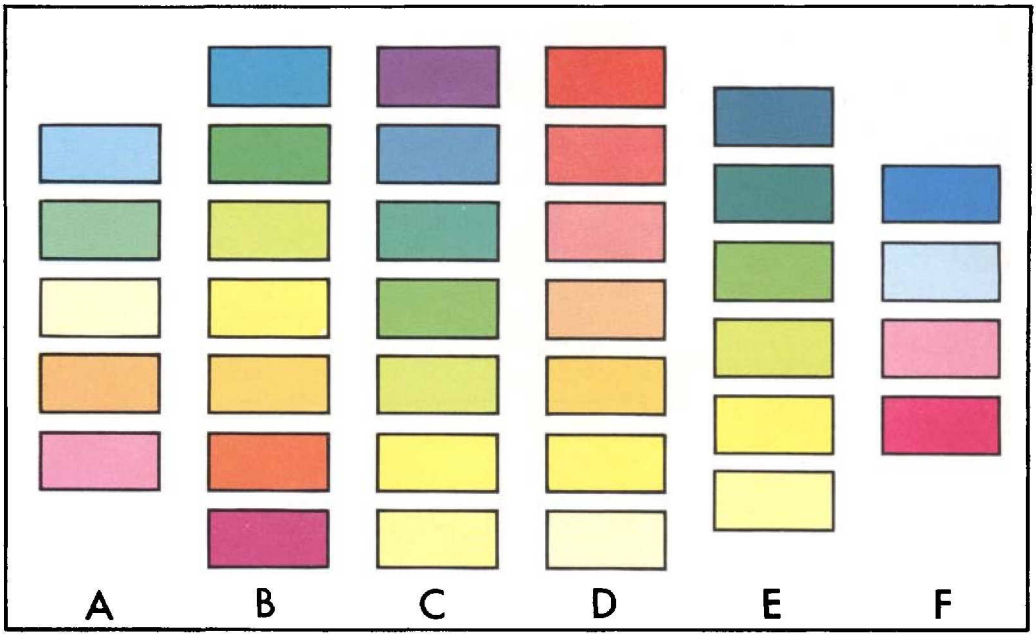


Fig. 2. Selected samples of color schemes used in the GE–50 series. These are necessarily approximations in that inks and papers used in the series differ slightly from those used here.

series, one of the most difficult problems has been obtaining proper balance of color; the selection has been aimed at showing the significant characteristics of the distribution without having some tones stand out inappropriately.

The GE–70 series is more recent and was initiated in 1974. At this time there is only one published map, but it has proven to be one of the most popular maps to be produced by the Census Bureau. It depicts the distribution of the 1970 U.S. population as a dot distribution. The map is printed in negative form with the U.S. land area appearing as a black background for white dots and is bordered by a field of deep blue. Originally published at 50 by 76 cm. (20 × 30 in.) at a 1:7,500,000 scale, it was not suitable for reduction to page size for this article; hence it is not illustrated here.

The second in this series is scheduled for publication in the near future. It is a two-variable map and will show the distribution of older Americans and year of maximum county population. This map and its two-variable technique are described below.

Two-Variable Maps

The increasing volume of data available that can be brought to bear on decision-making problems also increases the problems of how best to present the data. Maps are tools which can be used quite effectively by analysts and policymakers, whose time is often at a premium, and by members of the general public. Consideration of decisionmaking uses of maps has prompted the Bureau to examine alternative mapmaking techniques as well as to update its production technology. In 1974 the use of color in cartography took on a new importance when the Director of the Bureau suggested that two choropleth maps of different variables be "crossed." The first of these color "cross-maps" resulted from the combination of two existing GE–50 series maps, Distribution of Older Americans in 1970 and Year of Maximum County Population (Fig. 3). Using the original artwork, the Bureau's

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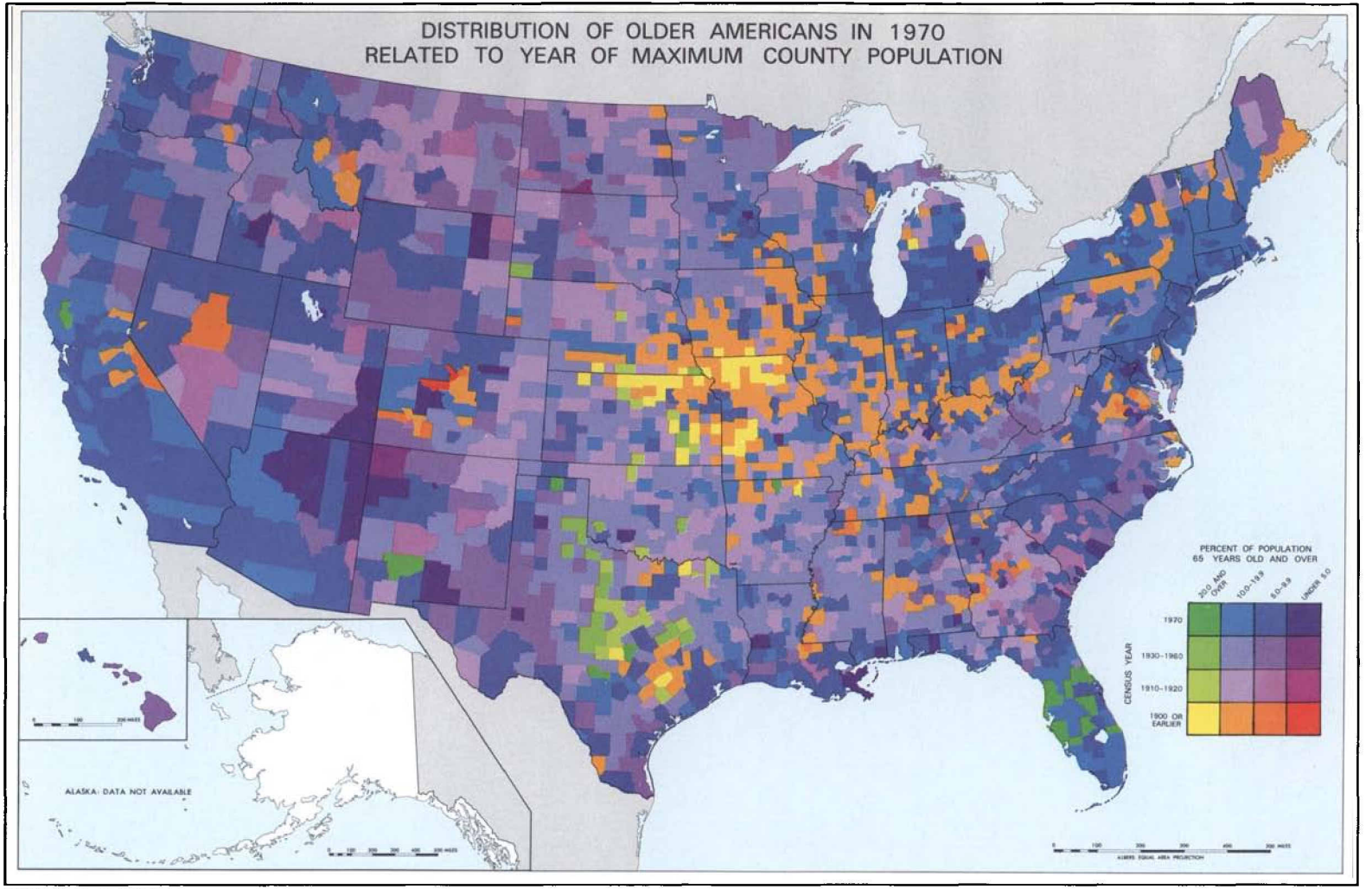


Figure 3.

staff chose a color scheme for each distribution, reduced the 1:5,000,000 color separations to the desired 1:7,500,000 size, and combined and screened the negatives into four color composites (red, blue, yellow, and black). Some of the potential of this technique was immediately obvious. The two-variable, cross-map technique was felt to be a means of conveying both the spatial distribution of two variables and of revealing information concerning the relationship between the variables.

Another pair of distributions, Percent High School Graduates and Per Capita Income (Figs. 4 and 5), were crossed to produce a two-variable map (Fig. 6), and this set of maps is included here to illustrate both the mechanics of the method and some of the advantages of the composite map. While the same information about distribution is indeed encoded in the separate maps of each variable, the cross-map technique appears to aid the reader in locating areas of special interest. A moment's study of the cross-map in Fig. 6 (interrelationship of education and income) reveals several interesting items: As expected, the colors of highest frequency of occurrence on the map suggest a positive overall relationship between the two variables. The appearance of a high percentage of high school graduates and low income (green areas) suggest that in these geographic areas income opportunities are not equal to those in other areas with comparable education levels. On the other hand, the areas in yellow show locations of low percentage of high school graduates and low income. Here another feature comes to light: the counties in yellow tend to cluster. The question becomes: Is there a socio-economic reason for the apparent clustering? In short, the two-variable cross-map highlights areas of special interest and shows them in relationship to the total picture.

Another example of the cross-map technique depicts the interrelationship of size of farm with farm income (Fig. 7). Among the interesting patterns on this map are the large farms with high income in the western United States and the small, high-income farms in proximity to urban areas. Of particular note is the yellow area of low income and small farm size in the Appalachian area. This is also the same general area of low education and low income seen in Fig. 6. This map and the ones in Figs. 4, 5, and 6 were produced by the Bureau's computer-assisted cartography system described below.

Research is currently underway to investigate how well cross-maps portray information. Among the facets under study are the effects on the reader of class interval selection, color choice, and the number of classes.

Micrographics

One of the major drawbacks to the use of maps as a tool in the decisionmaking process is the time and money required to produce them. The resources required to produce a two-variable cross-map are at least twice those of a single-variable map because, in effect, two separate maps must be produced first and then combined. Hence the cross-maps involve the total cartographic problems of producing two maps as well as the additional problems of combining them. Conventional procedures used at the Bureau of the Census for the preparation of choropleth maps involve the use of deep-etch peelcoats to make open-window color separations in negative form. If a map is to show data by counties, a separate peelcoat is prepared for each class interval.

The Bureau has developed an approach which significantly reduces both the cost and the time needed in statistical map preparation. It uses both automation and conventional techniques in a total system where each technique complements the other.[2] In this system the computer is used to process the data and to prepare the open-window negatives on a computer-output-on-microfilm (COM) device. This micrographics system has reduced the cost of preparing page-size color separations of U.S. county data to a fraction of the cost of the conventional procedure. The time required to prepare the separations has been correspondingly reduced from weeks

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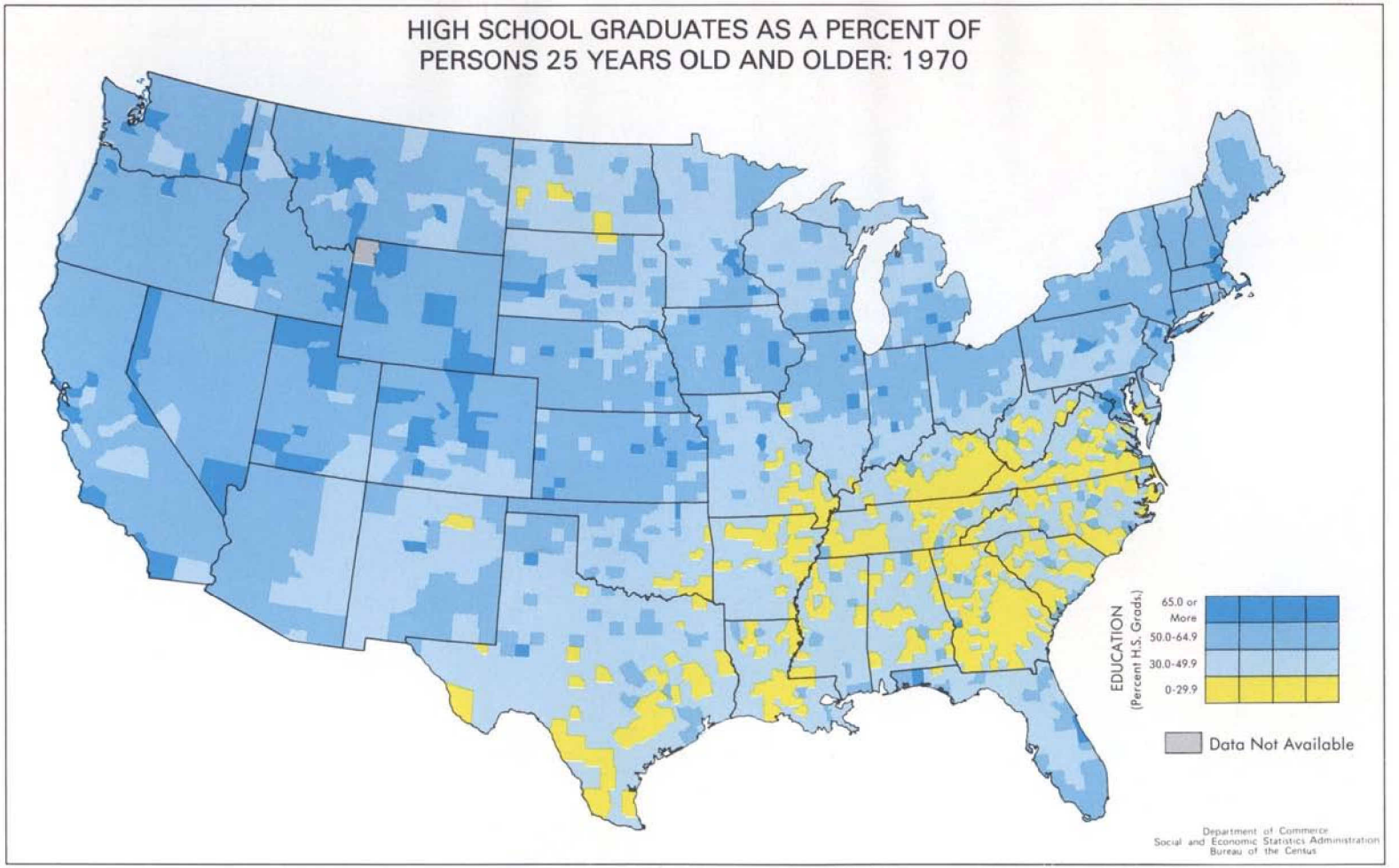


Figure 4.

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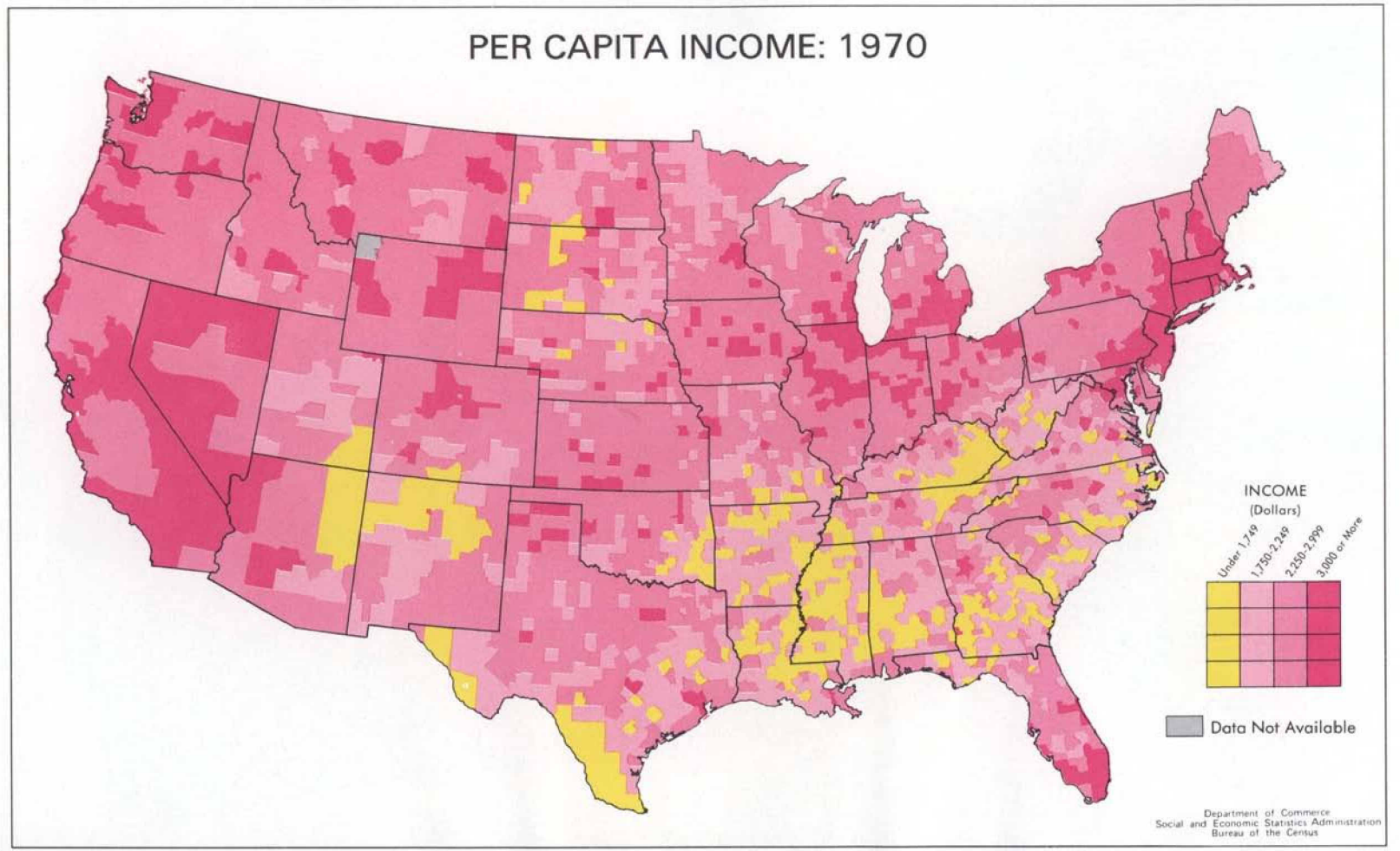


Figure 5.

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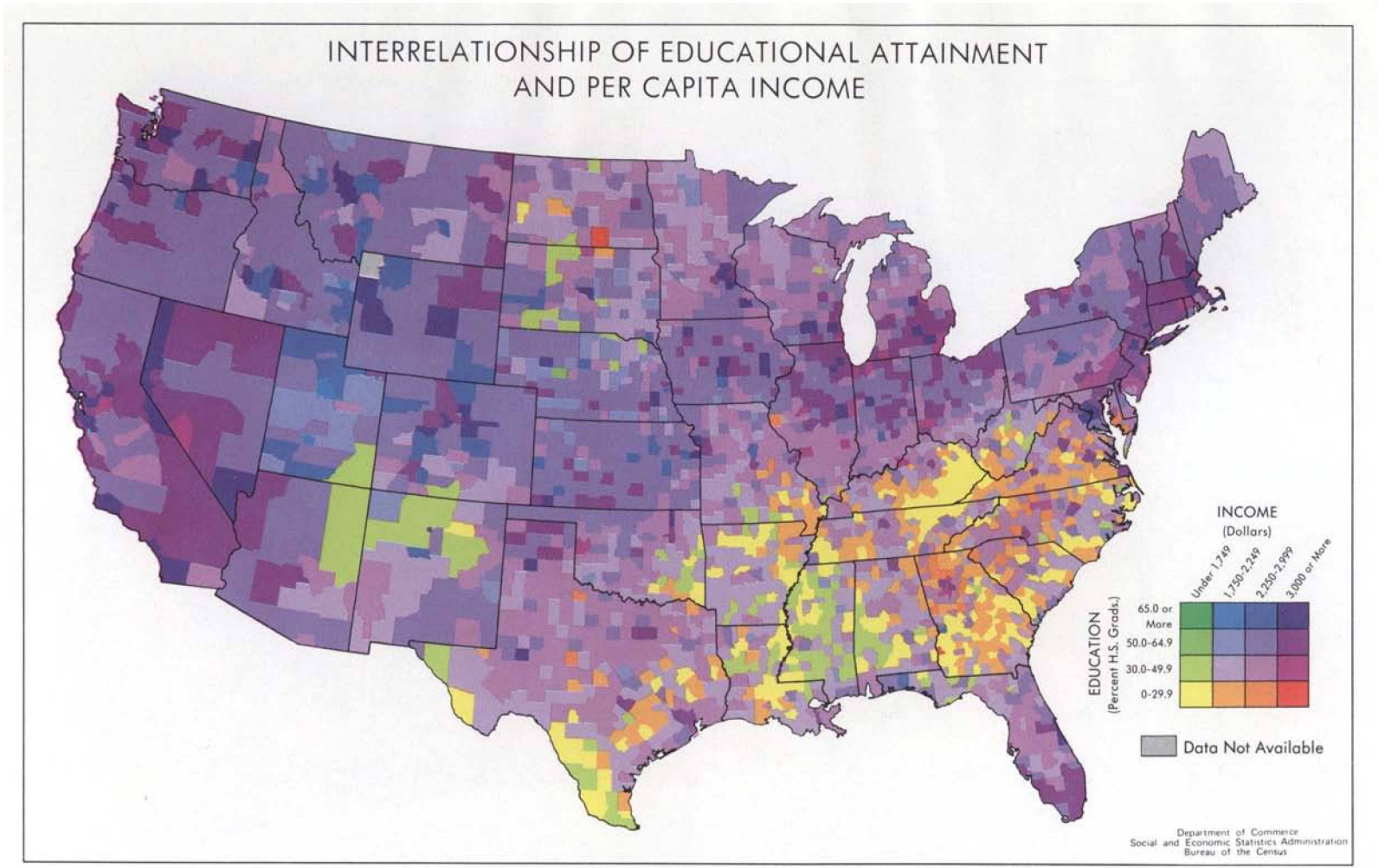


Figure 6.

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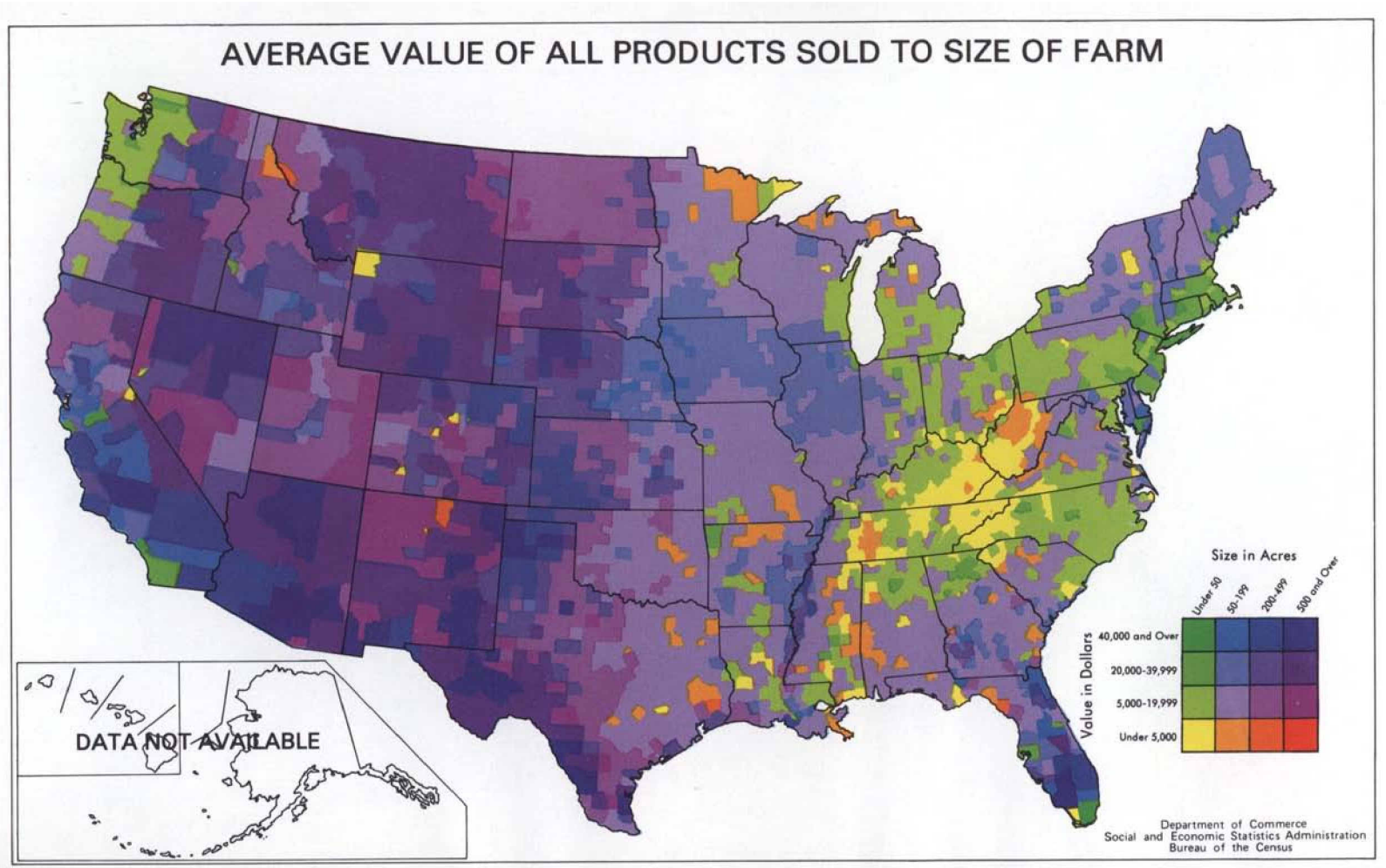


Figure 7.

to days or, in some instances, hours.

Figure 8 shows a diagram of the overall system. The variables to be mapped are extracted from a data base along with the codes of the geographic areas to be mapped, which in this case are the counties of the United States. A computer program then classes the extracted data and outputs an intermediate machine readable file. The other part of the system takes a file containing the description of the county boundaries in digital form, e.g., a series of x-y coordinates, and processes this boundary description to determine the position and size of each county as it is to be plotted on the map. This program outputs an intermediate file containing, for each county, a geographic code and a series of coordinates which will drive the plotting device to "paint" the image of the county.

Both intermediate files are processed through a program that uses the common geographic codes in the two intermediate files to match the class interval for each county with the county's image. The match program outputs to tape the appropriate commands to the plotting device to cause the county images to be drawn on the proper separation. The tape is sent to the COM unit where the commands are read and used to draw images on a CRT (cathode ray tube). The picture on the CRT is photographed by a camera containing 35mm film. The result is a 35mm open-window negative image.

The film is then sent to the Bureau's Geographic Operations Branch in Jeffersonville, Indiana, where the images are conventionally enlarged to the desired size and are composited and screened into color composite negatives. A color proof is made from the composites, and if acceptable, the composite negatives are sent to the printer to be used to make the press plates. The total time per map for the whole operation includes about five minutes CPU time on the Bureau's Univac 1108 computer (three minutes for image processing, two minutes for matching the class intervals to counties), five minutes of COM plotter time, one man-day of conventional processing in the photographic and cartographic sections, and variable amounts of time for the final printing operation since printing is done commercially.

The system allows for considerably increased speed, but only in producing maps of areas for which both the data and the boundary description information exist in machine readable form. Both of these conditions existed when the Census Bureau's Agriculture Division wanted to publish a graphic summary of the 1969 Census of Agriculture.[3] The system was used to produce over 60 color choropleth maps in just three months time. The preparation of this number of maps and the publication of this volume proved that the system was indeed a production system.

The Urban Atlases

The micrographics technique proved to be so successful that the Bureau decided to apply it to produce a major series of atlases.[4] The purpose of the Urban Atlas project was to produce atlases depicting selected socio-economic characteristics by census tract for Standard Metropolitan Statistical Areas (SMSA's). The digital descriptions of the census tract boundaries in 247 SMSA's were prepared, using a laser line-following digitizer. This work, done under contract through Lawrence Berkeley Laboratory, California, resulted in digital data descriptions which can be used for many mapping purposes. Due to funds limitations, atlases for only the 65 SMSA's over 500,000 population were scheduled for publication. In each of the 65 atlases, 11 variables are mapped individually and a 12th map is a cross of the education and income maps. Each map is 38 by 48 cm. (15 by 19 in.). A separate map shows the census tract boundaries and tract identification numbers for comparison with detailed data published in the census tract volumes. In a 14-month period, over 5,000 color separations for 780 maps were produced, and to date 16 SMSA atlases have been completed and released.

Figures 9, 10, 11, and 12 are reduced versions of the maps depicting median

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MICROGRAPHIC MAPPING AT THE CENSUS BUREAU

Translation of Census data files and digitized boundary files to maps via computer output on microfilm.

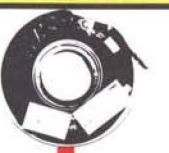
DATA FILE
This file contains a geographic code and statistical information for each area to be mapped.



CONTROL CARDS
These cards specify which statistic is to be mapped and the class intervals of the statistic. Up to six different maps may be produced at one time.



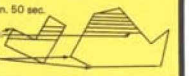
OUTLINE FILE (DIGITIZED BOUNDARY FILE)
This file contains a geographic code and a series of coordinates describing the boundary of each area to be mapped.



EXTRACTOR
Under control of the control cards, this operation extracts the specified statistic from the data file for each area, codes it to the correct class interval, and outputs the class code with the geographic code to an intermediate file.

SCAN IMAGE
This program takes the outline file coordinates for each geographic area and produces a file of scan lines. The scan lines are the beginning and end point coordinates of each sweep needed by the microfilm plotter's electron beam to "paint" the geographic area on the film. The geographic code for each area along with its associated scan coordinates are output to an intermediate file. Executing this program can be expensive in computer time when thousands of geographic areas are being scanned. Therefore, this intermediate file is usually stored and used again when the same area is to be mapped.

Normal run time: 3 min. 50 sec.



MATCH
This program uses the geographic codes in the files from the extractor and scan programs to match and assign a class code to the scan image records for each geographic area. The resulting records are then sorted by class code. Finally, the program outputs the image by class to a tape for the microfilm plotter. All the images for a class are plotted on one frame.

Normal run time: 2 min.

FR 80 COMPUTER OUTPUT ON MICROFILM DEVICE
This device reads the tape from the match program and produces the 35mm microfilm. The device is capable of addressing up to 16 thousand points along each axis. It plots the microfilm frames at approximately three a minute.

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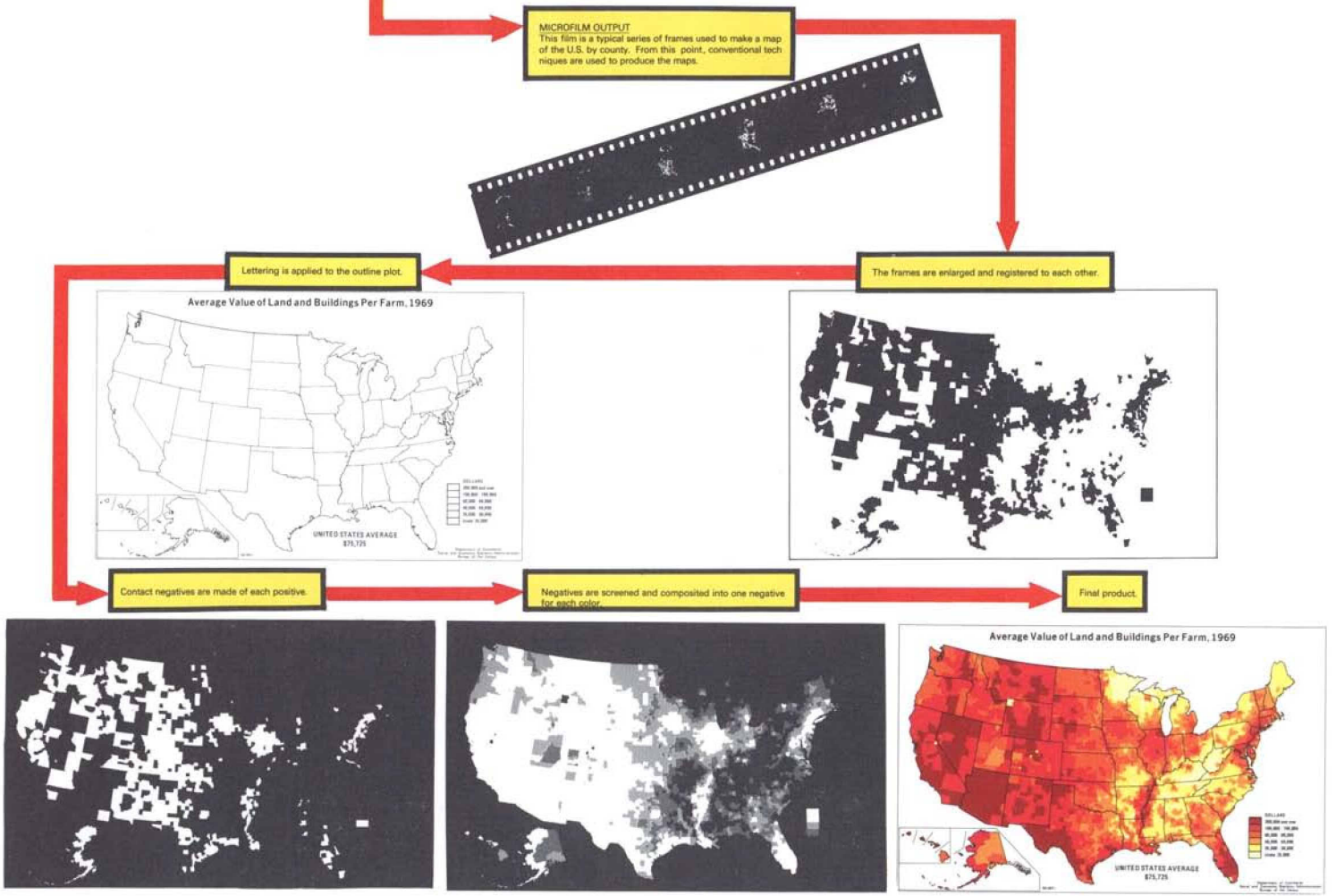


Figure 8.

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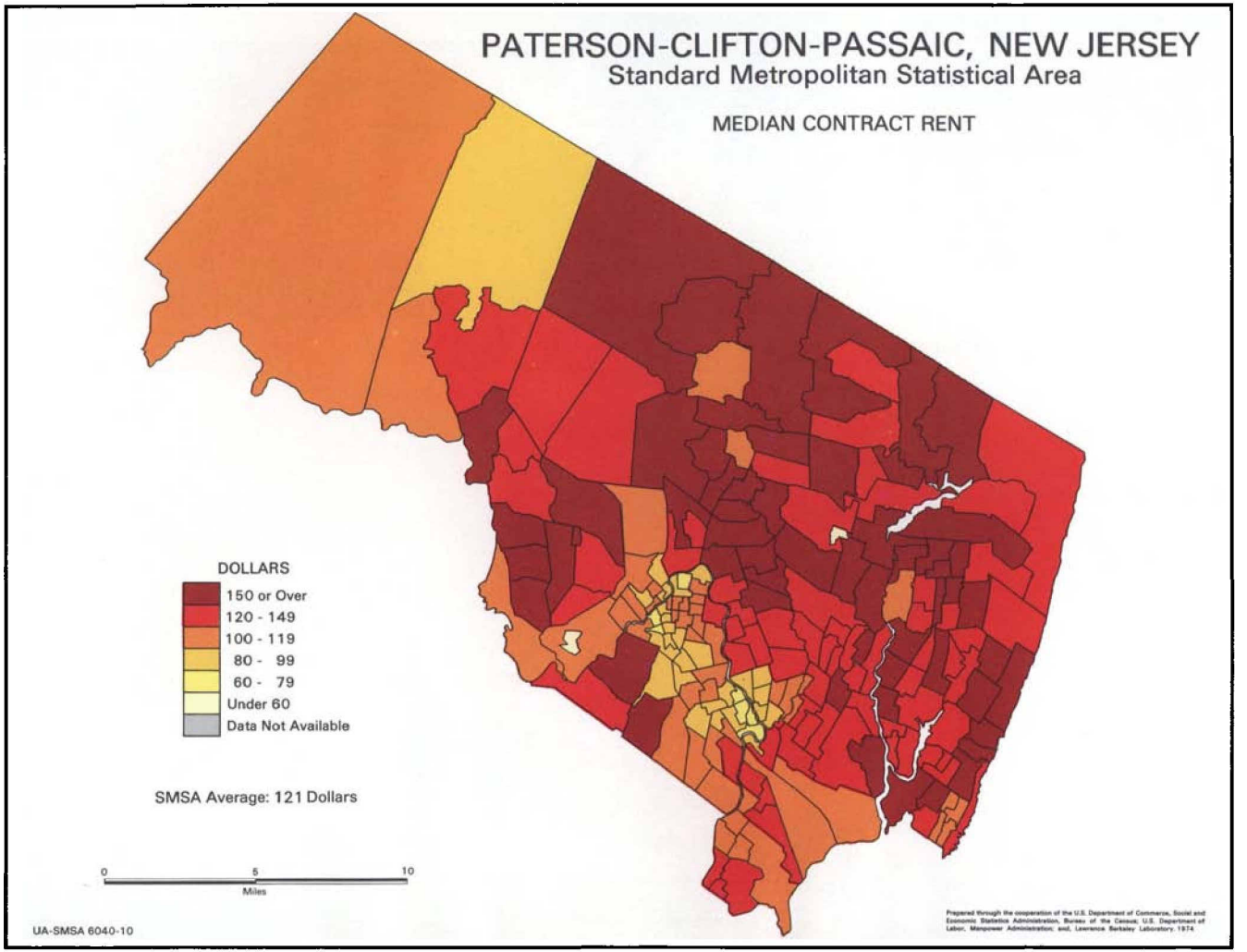


Figure 9.

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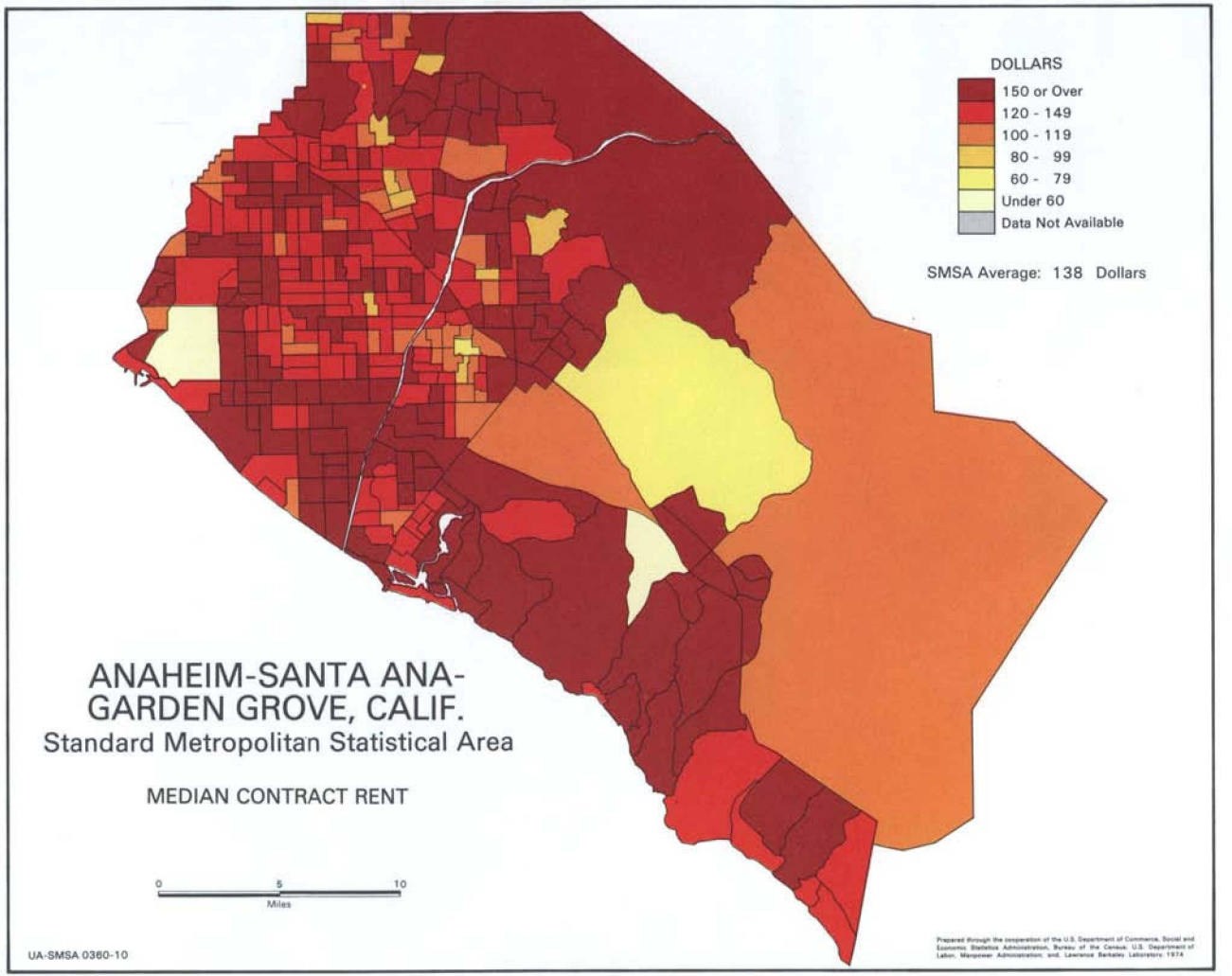


Figure 10.

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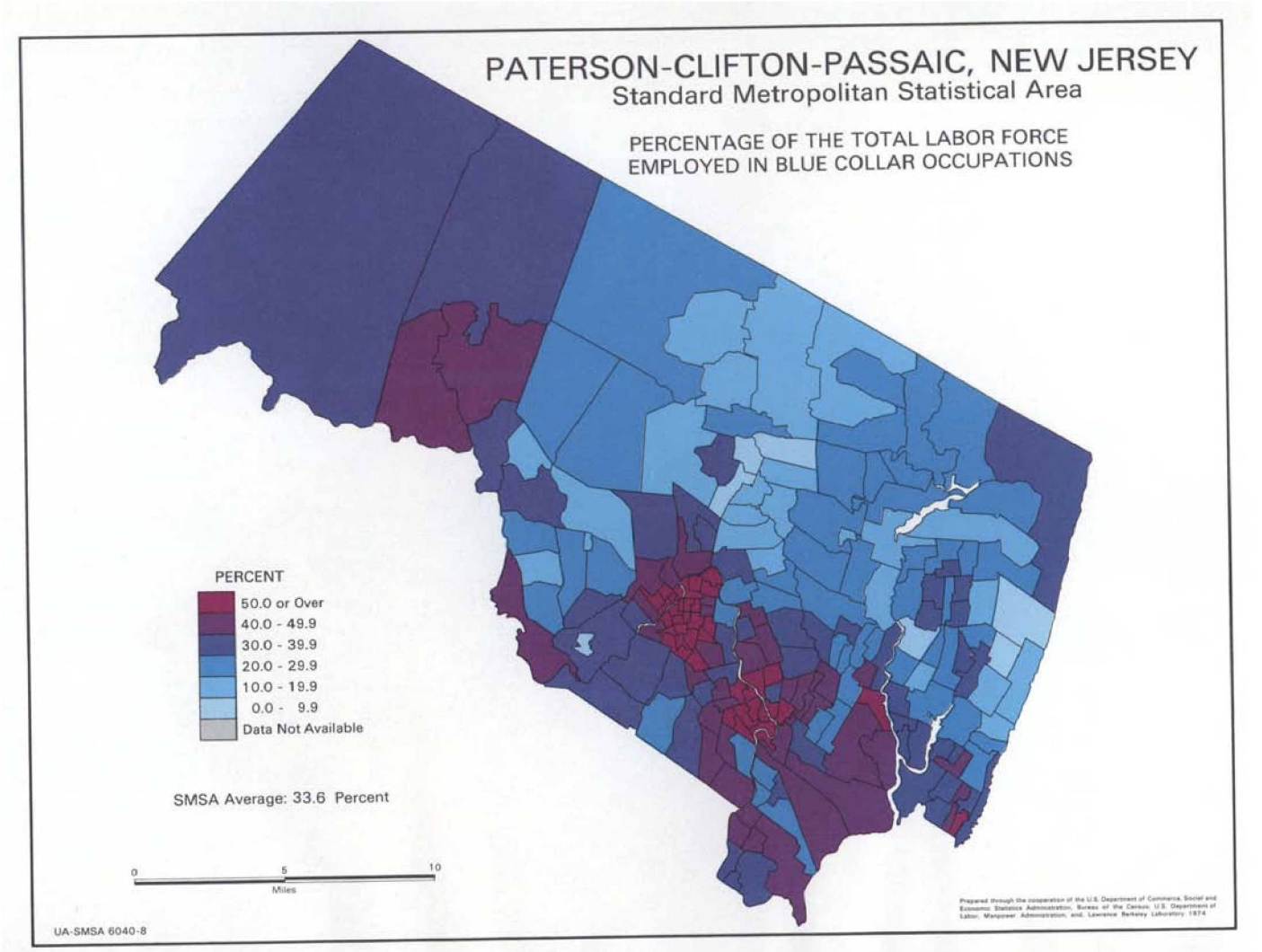


Figure 11.

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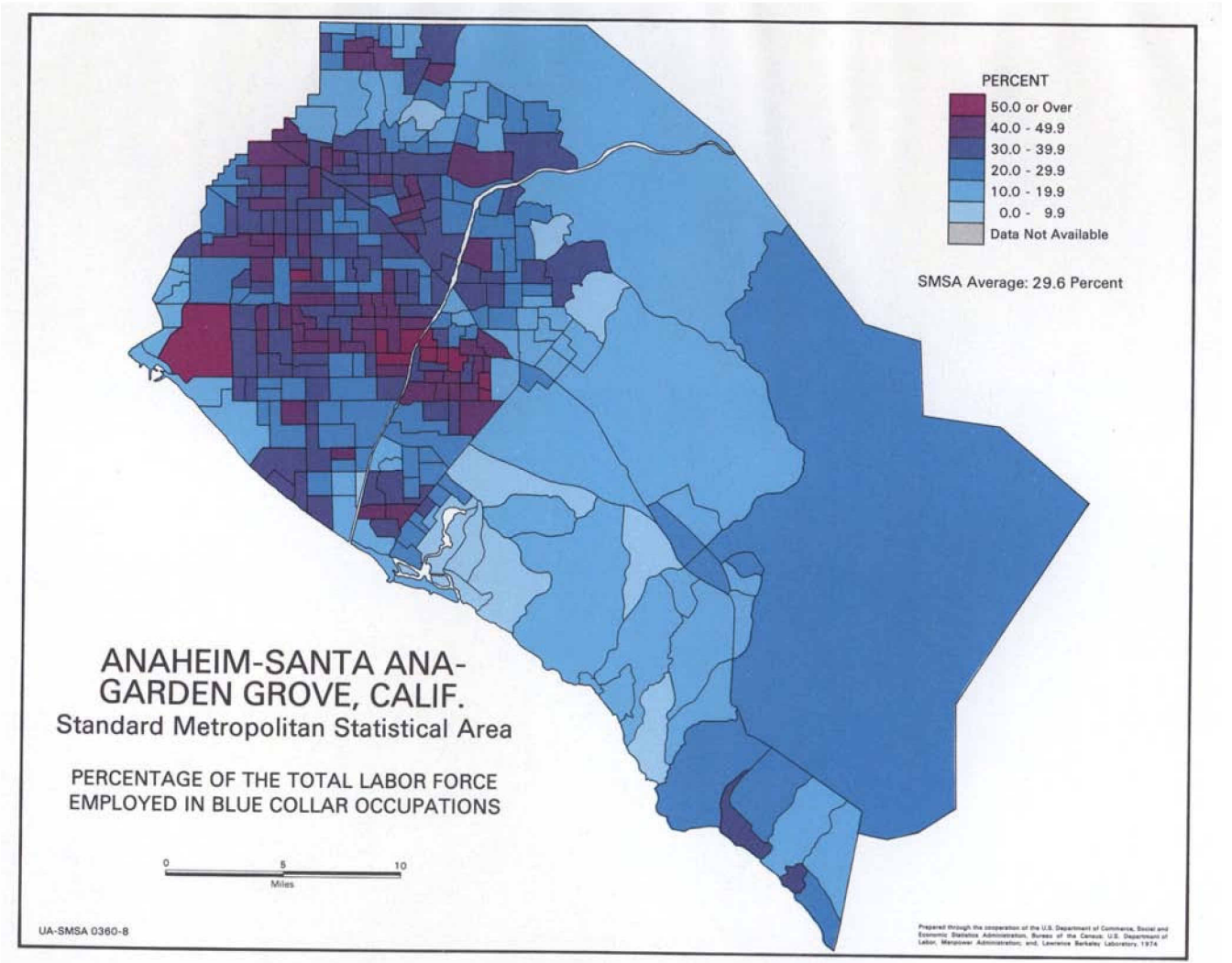


Figure 12.

contract rent and percent blue collar workers from the atlases of Paterson-Clifton-Passaic (New Jersey) and Anaheim-Santa Ana-Garden Grove (California).[5] Class intervals for the series were selected by analyzing the data for all the census tracts in the United States. The use of a standard legend for each map throughout the series was chosen to facilitate the comparison of distributions for different SMSA's. In the examples shown, one can readily compare the distribution of tracts of, say, 50% and over of total labor force in blue collar in the Paterson SMSA to the same class of data in Anaheim. Within SMSA, comparisons between distributions are easily made, for example, blue collar workers tend to live in low-rent districts. It should be kept in mind that these are merely reduced versions of the atlas sheets which were made through the micrographics system.

Other Projects in Progress

A wide variety of color thematic mapmaking activities are underway at the U.S. Bureau of the Census. They include the extension of both conventional techniques and automated methods into new ways of portraying data on maps. Under study are the uses of color and class interval choice in the GE–50, GE–70, and Urban Atlases series. The sole objective of the Bureau's efforts into better and faster mapping techniques is to provide the public and the policymaker improved understanding of the massive amount of data collected by the U.S. Census.

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APPENDIX

A listing of the maps in the GE–50 Series. Some maps are now out of print.

No. 1—Population Distribution, Urban and Rural, in the United States: 1960
No. 2—Standard Metropolitan Statistical Areas of the United States and Puerto Rico: 1963
No. 3—Congressional Districts for the 89th Congress (1965–1966)
No. 4—Housing Built Before and After 1950 in Standard Metropolitan Statistical Areas of the United States: 1960
No. 5—Housing Owned and Rented in Standard Metropolitan Statistical Areas of the United States: 1960
No. 6—Families With Incomes Under $3,000 in 1959, by Counties of the United States: 1960
No. 7—Older Americans in the United States, by Counties: 1960
No. 8—Per Capita Money Income for 1959, by Counties of the United States
No. 9—Population With High School Education or More, by Counties of the United States: 1960
No. 10—Youths of 16 and 17 Years of Age in School, by Counties in the United States: 1960
No. 11—Congressional Districts for the 90th Congress (1967–1968)
No. 12—Employment in Manufacturing, by Counties of the United States: 1960
No. 13—Employment in Professional, Technical, and Kindred Occupations, by Counties of the United States: 1960
No. 14—American Indians in the United States: 1960
No. 15—Japanese and Chinese in the United States: 1960
No. 16—Negro Population as Percent of Total Population, by Counties of the United States: 1960
No. 17—Standard Metropolitan Statistical Areas in the United States and Puerto Rico: 1967
No. 18—Retail Trade in the United States: 1963
No. 19—Sales of Retail Shopping Goods Stores for Selected Standard Metropolitan Statistical Areas in the United States: 1963
No. 20—Wholesale Trade in the United States: 1963

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No. 21—Hotel-Motel Receipts in the United States: 1963
No. 22—Manufacturing in the United States: 1963
No. 23—Mineral Industries in the United States: 1963
No. 24—Families in Poverty Areas for Selected Cities of the United States: 1960
No. 25—Congressional Districts for the 91st Congress (Including Redistricting Actions to August 1, 1968)
No. 26—Shipments of Commodities by Manufacturers in the Conterminous United States, OUTFLOW from Census Geographic Divisions: 1963
No. 27—Shipments of Commodities by Manufacturers in the Conterminous United States, INFLOW to Census Geographic Divisions: 1963
No. 28—Housing Construction Authorized in Permit Issuing Pieces, by Selected Standard Metropolitan Statistical Areas of the United States: 1964–1967
No. 29—Size of Farms, by Counties of the United States: 1964
No. 30—Percent Change in Size of Farms, by Counties of the United States: 1940–1964
No. 31—Value of Farm Products Sold per Acre of Land in Farms, by Counties of the United States: 1964
No. 32—Percent Change in the Value of Farm Products Sold per Acre of Land in Farms, by Counties of the United States: 1939–1964
No. 33—Corn Yield per Acre, by Counties of the United States: 1964
No. 34—Percent Change in Corn Yield per Acre, by Counties of the United States: 1939–1964
No. 35—Congressional Districts for the 92nd Congress (Includes Redistricting Actions to August 1, 1970)
No. 36—Older Americans by Counties of the United States: 1970
No. 37—Year of Maximum Population by Counties of the United States
No. 38—Population Density by Counties of the United States: 1970
No. 39—Percent of Population Urban by Counties of the United States: 1970
No. 40—Standard Metropolitan Statistical Areas of the United States and Puerto Rico: 1970
No. 41—Percent Change in Population by Counties of the United States: 1960–1970
No. 42—Population Trends by Counties of the United States: 1940–1970
No. 43—1970 Population as a Percent of Maximum Population by Counties of the United States
No. 44—Net Migration by Counties of the United States: 1960–1970
No. 45—Population Distribution, Urban and Rural, in the United States: 1970
No. 46—Congressional Districts for the 93rd Congress
No. 47—Number of Negro Persons, by Counties of the United States: 1970
No. 48—Negro Population as a Percent of Total Population, by Counties of the United States: 1970
No. 49—Number of American Indians, by Counties of the United States: 1970
No. 50—Number of Chinese, by Counties of the United States: 1970
No. 51—Number of Japanese, by Counties of the United States: 1970
No. 52—Number of Persons of Spanish Origin, by Counties of the United States: 1970
No. 53—Number of Filipinos by Counties of the United States: 1970
No. 54—Percent Change in the Negro Population, by Counties of the United States: 1960–1970
No. 55—Standard Metropolitan Statistical Areas; Area Defined by Office of Management and Budget: January 1, 1974
No. 56—Median Family Income for 1969, by Counties of the United States: 1970
No. 57—Per Capita Money Income for 1969, by Counties of the United States
No. 58—Families Below the Low-Income Level in 1969 by Counties of the United States
No. 59—Number of Owner Occupied Housing Units by Counties of the United States: 1970
No. 60—Number of Renter Occupied Housing Units by Counties of the United States: 1970
No. 61—Spanish Origin Persons as a Percent of Total Population by Counties of the United States: 1970
No. 62—Congressional Districts for the 94th Congress

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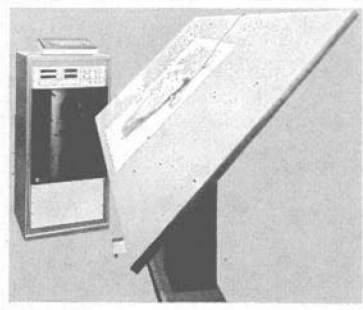
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